Solutions

Maths

Q1. Let $f:[0,3]\to\mathbb{R}$ be a function defined by $f(x)=2x^3-9x^2+12x+1$ Let M and m be the absolute maximum and absolute minimum values of $f(x)$ on $[0,3]$, respectively. If (x_0,y_0) is the point of inflection on the graph of $y=f(x)$, then the value of $M+m+x_0+y_0$ is equal to:

(A) 16 (B) 18 (C) 20 (D) 22

Ans. B

Sol. $f(x)=2x^3-9x^2+12x+1$: $f'(x)=6x^2-18x+12=6(x-1)(x-2)$

Critical points in $[0,3]$ are $x=1$ and $x=2$.

$f(0)=1$, $f(1)=6$; $f(2)=5$; $f(3)=10$

Thus, absolute maximum value $M=10$ and absolute minimum value $m=1$.

$f''(x)=12x-18$; Setting $f''(x)=0 \implies x_0=\dfrac{3}{2}$.

Since $f''(x)$ changes sign across $x=\dfrac{3}{2}$, (x_0,y_0) is a point of inflection.

$$y_0=f\left(\frac{3}{2}\right)=2\left(\frac{27}{8}\right)-9\left(\frac{9}{4}\right)+12\left(\frac{3}{2}\right)+1=\frac{11}{2}$$

$$M+m+x_0+y_0=10+1+\frac{3}{2}+\frac{11}{2}=11+7=18$$

Q2. Let $L=\lim\limits_{x\to0^+}(\sin x)^x$. Consider the function $f:\left[0,\dfrac{\pi}{2}\right]\to\mathbb{R}$ defined by

$$f(x)=\begin{cases}(\sin x)^x, & x\in\left(0,\frac{\pi}{2}\right]\\ L, & x=0\end{cases},$$

If $g(x)=f(x)-x-\dfrac{1}{2}$, then the equation $g(x)=0$ has:

(A) no root in the interval $\left(0,\dfrac{\pi}{2}\right)$ (B) no root in the interval $\left[0,\dfrac{\pi}{2}\right]$

(C) exactly two roots in the interval $\left(\dfrac{\pi}{2},\pi\right)$ (D) at least one root in the interval $\left(0,\dfrac{\pi}{2}\right)$

Ans. D

Sol. $\ln L=\lim\limits_{x\to0^+}x\ln(\sin x)=\lim\limits_{x\to0^+}\dfrac{\ln(\sin x)}{\frac{1}{x}}$

$$\ln L=\lim_{x\to0^+}\frac{\cot x}{-\frac{1}{x^2}}=-\lim_{x\to0^+}\frac{x^2}{\tan x}=0 \implies L=e^0=1$$

Thus, $f(0)=1$, making $f(x)$ continuous at $x=0$ and consequently continuous on the closed interval $\left[0,\dfrac{\pi}{2}\right]$.

The function $g(x) = f(x) - x - \frac{1}{2}$ is also continuous on $\left[0, \frac{\pi}{2}\right]$.

$g(0) = f(0) - 0 - \frac{1}{2} = 1 - \frac{1}{2} = \frac{1}{2} > 0$

$g\left(\frac{\pi}{2}\right) = f\left(\frac{\pi}{2}\right) - \frac{\pi}{2} - \frac{1}{2} = (\sin\frac{\pi}{2})^{\pi/2} - \frac{\pi}{2} - \frac{1}{2} = 1 - \frac{\pi}{2} - \frac{1}{2} = \frac{1-\pi}{2} < 0$

Since $g(0) \cdot g\left(\frac{\pi}{2}\right) < 0$ and $g(x)$ is continuous on $\left[0, \frac{\pi}{2}\right]$, by the Intermediate Value Theorem, there exists at least one $c \in \left(0, \frac{\pi}{2}\right)$ such that $g(c) = 0$.

Q3. Let $y = f(x)$ be a continuous function defined on $\mathbb{R}$. For all $x \neq 2$, the slope of the tangent to the curve $y = f(x)$ at the point $(x, f(x))$ is given by $g(x) = \frac{x^2 - 4}{|x - 2|}$. The type of discontinuity possessed by the slope function $g(x)$ at $x = 2$ is:

(A) Removable discontinuity (B) Jump discontinuity (C) Infinite discontinuity

(D) Oscillatory discontinuity

Ans. B

Sol. $\text{RHL} = \lim\limits_{x \to 2^+} g(x) = \lim\limits_{x \to 2^+} \frac{(x-2)(x+2)}{x-2} = \lim\limits_{x \to 2^+} (x+2) = 4$

$\text{LHL} = \lim\limits_{x \to 2^-} g(x) = \lim\limits_{x \to 2^-} \frac{(x-2)(x+2)}{-(x-2)} = \lim\limits_{x \to 2^-} -(x+2) = -4$

Since both LHL and RHL exist finitely but are not equal (LHL $\neq$ RHL), $g(x)$ has a jump discontinuity (non-removable discontinuity of the first kind) at $x = 2$.

Q4. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function satisfying $f(x + y) = f(x) + f(y) + 3xy(x + y)$ for all $x, y \in \mathbb{R}$. If $f'(0) = 1$ and $g(x)$ is the inverse function of $f(x)$, then the value of $g'(2)$ is equal to:

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 4 (D) $\frac{1}{7}$

Ans. A

Sol. Using the first principle of differentiation for $f(x)$:

$f'(x) = \lim\limits_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim\limits_{h \to 0} \frac{f(h) + 3xh(x+h)}{h}$

$f'(x) = \lim\limits_{h \to 0} \frac{f(h)}{h} + \lim\limits_{h \to 0} 3x(x+h) = f'(0) + 3x^2 = 3x^2 + 1$

Integrating $f'(x) = 3x^2 + 1$ with respect to x and using $f(0) = 0$ we get: $f(x) = x^3 + x$

Since $g(x) = f^{-1}(x)$, we have $f(g(x)) = x$.

$g'(x) = \frac{1}{f'(g(x))}$

Let $f(x) = 2 \implies x^3 + x = 2 \implies x = 1$, so $g(2) = 1$.

Now, $g'(2) = \frac{1}{f'(1)}$.

Since $f'(1) = 3(1)^2 + 1 = 4$, we get: $g'(2) = \frac{1}{4}$

Q5. Let $f:[1,e]\to\mathbb{R}$ be a continuous function on $[1,e]$ and differentiable on $(1,e)$ such that $f(1)=0$ and $f(e)=e$.

Consider the function $g:[1,\infty)\to\mathbb{R}$ defined by
$g(x)=x\ln x-(x-1)e^{x-1}$

Which of the following options is **CORRECT** ?

(A) $g(x)>0$ for all $x>1$, and there exists at least one $c\in(1,e)$ such that $f'(c)=1+\ln c$.

(B) $g(x)<0$ for all $x>1$, and there exists at least one $c\in(1,e)$ such that $f'(c)=1+\ln c$.

(C) $g(x)>0$ for all $x>1$, and there exists at least one $c\in(1,e)$ such that $f'(c)=\ln c$.

(D) $g(x)<0$ for all $x>1$, and there exists at least one $c\in(1,e)$ such that $f'(c)=\ln c$.

Ans. B

Sol. $g(x)=x\ln x-(x-1)e^{x-1}$ for $x\geq 1$.
$g'(x)=1+\ln x-e^{x-1}-(x-1)e^{x-1}=1+\ln x-xe^{x-1}$
Let $k(x)=g'(x)=1+\ln x-xe^{x-1}$.
Differentiating $k(x)$:
$k'(x)=\dfrac{1}{x}-e^{x-1}-xe^{x-1}=\dfrac{1}{x}-(x+1)e^{x-1}$
For $x>1$, we have $\dfrac{1}{x}<1$ and $(x+1)e^{x-1}>2$.
Thus, $k'(x)<0$ for all $x>1$.
Since $k'(x)<0$, $k(x)$ is strictly decreasing on $[1,\infty)$.
Since $k(1)=1+0-1=0$, it follows that $g'(x)=k(x)<0$ for all $x>1$.
Since $g'(x)<0$, $g(x)$ is strictly decreasing on $[1,\infty)$.
Given $g(1)=0$, we conclude that $g(x)<0$ for all $x>1$.

Define another function $h(x)=f(x)-x\ln x$ on $[1,e]$.
$h(1)=f(1)-1\ln 1=0$; $h(e)=f(e)-e\ln e=e-e=0$

Since $h(x)$ is continuous on $[1,e]$ and differentiable on $(1,e)$, by Rolle's Theorem, there exists at least one $c\in(1,e)$ such that $h'(c)=0$.
$h'(x)=f'(x)-(1+\ln x)$
$h'(c)=0\implies f'(c)=1+\ln c$

Q6. Let $y=y(x)$ be a differentiable function defined implicitly by the determinant equation $\begin{vmatrix} y & x^2 \\ x & y \end{vmatrix}=3$ If $y(1)=2$, then the value of $\left.\dfrac{dy}{dx}\right|_{x=1}$ is equal to:

(A) $\dfrac{3}{4}$ (B) $\dfrac{4}{3}$ (C) $\dfrac{1}{2}$ (D) $\dfrac{2}{3}$

Ans. A

Sol. $\begin{vmatrix} y & x^2 \\ x & y \end{vmatrix}=3$
$y^2-x^3=3$

$$2y\frac{dy}{dx} - 3x^2 = 0 \implies \frac{dy}{dx} = \frac{3x^2}{2y}$$

At $x = 1$ and $y = 2$:

$$\left.\frac{dy}{dx}\right|_{(1,2)} = \frac{3(1)^2}{2(2)} = \frac{3}{4}$$

Q7. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \max\{|x-1|, x^2 - 2x\}$ Then, the function $f(x)$ is:

(A) continuous everywhere and non-differentiable at exactly 2 points

(B) discontinuous at 1 point and non-differentiable at 3 points

(C) continuous everywhere and non-differentiable at exactly 3 points

(D) continuous everywhere and non-differentiable at exactly 1 point

Ans. C

Sol. Since $g(x) = |x-1|$ and $h(x) = x^2 - 2x$ are continuous on $\mathbb{R}$, $f(x) = \max\{g(x), h(x)\}$ is continuous everywhere on $\mathbb{R}$.

$$f(x) = \begin{cases} x^2 - 2x, & x < \frac{1-\sqrt{5}}{2} \\ 1 - x, & \frac{1-\sqrt{5}}{2} \le x < 1 \\ x - 1, & 1 \le x < \frac{3+\sqrt{5}}{2} \\ x^2 - 2x, & x \ge \frac{3+\sqrt{5}}{2} \end{cases}$$

Checking differentiability at critical points:

At $x = \dfrac{1-\sqrt{5}}{2}$, $f'(x^-) = 2x - 2 \neq -1 = f'(x^+)$.

At $x = 1$, $f'(x^-) = -1 \neq 1 = f'(x^+)$.

At $x = \dfrac{3+\sqrt{5}}{2}$, $f'(x^-) = 1 \neq 2x - 2 = f'(x^+)$.

Hence, $f(x)$ is continuous everywhere and non-differentiable at exactly 3 points: $x = \dfrac{1-\sqrt{5}}{2}$, $x = 1$, and $x = \dfrac{3+\sqrt{5}}{2}$.

Q8. Let $f(x) = x^3 - 3x + 2$. If $P(x_0, y_0)$ is the point corresponding to the local minimum of $f(x)$, then the shortest distance from the point P to the line $y = x - 4$ is equal to:

(A) $\sqrt{2}$ (B) $\dfrac{3}{\sqrt{2}}$ (C) $2\sqrt{2}$ (D) $\dfrac{5}{\sqrt{2}}$

Ans. B

Sol. $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$

Setting $f'(x) = 0$ gives $x = 1$ and $x = -1$.

Since $f''(x) = 6x$, $f''(1) = 6 > 0$, so $x = 1$ is the point of local minimum.

At $x = 1$, $f(1) = 1^3 - 3(1) + 2 = 0$.

Thus, the local minimum point is $P(1,0)$.

The line is given by $x - y - 4 = 0$.

The perpendicular distance from $P(1,0)$ to the line is:

$$d = \frac{|1 - 0 - 4|}{\sqrt{1^2 + (-1)^2}} = \frac{3}{\sqrt{2}}$$

Q9. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = [x] + \{x\}^2$, where $[x]$ and $\{x\}$ denote the greatest integer function and the fractional part function of x, respectively. Let $g(x) = |2x - 1|$. The total number of points in the open interval $(0, 2)$ at which the composite function $h(x) = (f \circ g)(x)$ is non-differentiable is:

(A) 1 (B) 4 (C) 3 (D) 2

Ans. D

Sol. For any $x \in \mathbb{R}$, let $n = [x] \in \mathbb{Z}$.

Then for $x \in [n, n+1)$, $f(x) = n + (x - n)^2$.

$\lim\limits_{x \to n^-} f(x) = (n-1) + 1^2 = n = f(n) = \lim\limits_{x \to n^+} f(x)$, so $f(x)$ is continuous everywhere.

$$f'(n^-) = \frac{d}{dx}\big((n-1) + (x - n + 1)^2\big)\Big|_{x=n} = 2.$$

$$f'(n^+) = \frac{d}{dx}\big(n + (x-n)^2\big)\Big|_{x=n} = 0.$$

Thus, $f(x)$ is non-differentiable at all integer points $k \in \mathbb{Z}$.

Now, consider $h(x) = f(g(x)) = f(|2x - 1|)$ for $x \in (0, 2)$.

1. At $x = 1/2$: $g(1/2) = 0$.
For $x \in (0, 1)$, $0 \le g(x) < 1$, which gives $[g(x)] = 0$.
Hence, $h(x) = (g(x))^2 = (2x - 1)^2$, which is a smooth polynomial on $(0, 1)$.
Thus, $h(x)$ is differentiable at $x = 1/2$.

2. At $x = 1$: $g(1) = 1 \in \mathbb{Z}^+$.
For $x \in (1/2, 1)$, $h(x) = (2x - 1)^2 \implies \lim\limits_{x \to 1^-} h'(x) = 4$.
For $x \in (1, 3/2)$, $g(x) \in (1, 2) \implies h(x) = 1 + (2x - 2)^2$

$\implies \lim\limits_{x \to 1^+} h'(x) = 0$.
Since LHD $\neq$ RHD at $x = 1$, $h(x)$ is non-differentiable at $x = 1$.

3. At $x = 3/2$: $g(3/2) = 2 \in \mathbb{Z}^+$.
For $x \in (1, 3/2)$, $h(x) = 1 + (2x - 2)^2 \implies \lim\limits_{x \to (3/2)^-} h'(x) = 4$.
For $x \in (3/2, 2)$, $g(x) \in (2, 3) \implies h(x) = 2 + (2x - 3)^2$

$\implies \lim\limits_{x\to(3/2)^+} h'(x) = 0.$

Since LHD $\neq$ RHD at $x = 3/2$, $h(x)$ is non-differentiable at $x = 3/2$.

Hence, $h(x)$ is non-differentiable at exactly 2 points in $(0, 2)$, which are $x = 1$ and $x = 3/2$.

Q10. If the curves $C_1 : y = x^3 + ax$ and $C_2 : y = k - \dfrac{x^2}{8}$ intersect orthogonally at a point with x-coordinate 1, then the number of distinct real roots of the equation $x^3 + ax - k = 0$ is equal to:

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. For $C_1 : y = x^3 + ax$, $\dfrac{dy}{dx} = 3x^2 + a$. At $x = 1$, slope $m_1 = 3 + a$.

For $C_2 : y = k - \dfrac{x^2}{8}$, $\dfrac{dy}{dx} = -\dfrac{x}{4}$. At $x = 1$, slope $m_2 = -\dfrac{1}{4}$.

Since the curves intersect orthogonally: $m_1 \cdot m_2 = -1 \implies (3 + a)\left(-\dfrac{1}{4}\right) = -1 \implies a = 1$

At the point of intersection $x = 1$, the y-coordinates must be equal: $y_1(1) = 1^3 + (1)(1) = 2$

$y_2(1) = k - \dfrac{1}{8} = 2 \implies k = \dfrac{17}{8}$

Now consider $f(x) = x^3 + ax - k = x^3 + x - \dfrac{17}{8}$.

$f'(x) = 3x^2 + 1 > 0 \quad \forall x \in \mathbb{R}$

Since $f(x)$ is strictly increasing on $\mathbb{R}$ and is continuous, by Intermediate Value Theorem and monotonicity, the equation $f(x) = 0$ has exactly 1 real root.

Q11. A point P moves along the negative y-axis away from the origin at a constant speed of 2 units/sec. From P, two tangents are drawn to the parabola $y = x^2$. If θ is the acute angle between these two tangents, then the rate of change of θ (in rad/s) at the instant when P is at $(0, -1)$ is equal to:

(A) $-\dfrac{4}{5}$ (B) $-\dfrac{2}{5}$ (C) $\dfrac{4}{5}$ (D) $\dfrac{3}{5}$

Ans. A

Sol. Let the coordinates of point P at any time t be $(0, -a)$, where $a > 0$.

Given that P moves away from the origin at 2 units/sec, we have $\dfrac{da}{dt} = 2$.

The equation of a tangent to the parabola $y = x^2$ with slope m is: $y = mx - \dfrac{m^2}{4}$

Since the tangent passes through $P(0, -a)$: $-a = -\dfrac{m^2}{4} \implies m^2 = 4a \implies m = \pm 2\sqrt{a}$

Let the slopes of the two tangents be $m_1 = 2\sqrt{a}$ *and* $m_2 = -2\sqrt{a}$.

The acute angle θ between the tangents satisfies: $\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\dfrac{4\sqrt{a}}{1 - 4a}\right|$

For $a = 1$, $1 - 4a = -3 < 0$, so for $a > \dfrac{1}{4}$: $\tan\theta = \dfrac{4\sqrt{a}}{4a - 1}$

Differentiating both sides with respect to t: $\sec^2\theta \dfrac{d\theta}{dt} = \dfrac{(4a-1)\cdot\frac{2}{\sqrt{a}} - 4\sqrt{a}\cdot 4}{(4a-1)^2} \cdot \dfrac{da}{dt}$

At the instant when P is at $(0, -1)$, we have $a = 1$: $\tan\theta = \dfrac{4}{3} \implies \sec^2\theta = 1 + \tan^2\theta = \dfrac{25}{9}$

Substituting $a = 1$, $\sec^2\theta = \dfrac{25}{9}$, and $\dfrac{da}{dt} = 2$:

$\dfrac{25}{9}\dfrac{d\theta}{dt} = \dfrac{(3)(2) - 16}{3^2} \cdot 2 = \dfrac{-10}{9} \cdot 2 = -\dfrac{20}{9}$

$\dfrac{d\theta}{dt} = -\dfrac{20}{25} = -\dfrac{4}{5}$ rad/s

Q12. Let m be the smallest positive integer such that the function

$$f(x) = \begin{cases} x^m \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$ is differentiable at $x = 0$.

A right circular cylinder of maximum volume is inscribed inside a right circular cone of base radius m units and height 3m units. The maximum volume of this inscribed cylinder (in cubic units) is:

(A) $\dfrac{16\pi}{9}$ (B) $\dfrac{32\pi}{9}$ (C) $\dfrac{8\pi}{3}$ (D) $\dfrac{64\pi}{27}$

Ans. B

Sol. For $f(x)$ to be differentiable at $x = 0$:

$f'(0) = \lim\limits_{h\to 0} \dfrac{f(h) - f(0)}{h} = \lim\limits_{h\to 0} \dfrac{h^m \sin\left(\frac{1}{h}\right) - 0}{h}$

$= \lim\limits_{h\to 0} h^{m-1} \sin\left(\dfrac{1}{h}\right)$

Since $\sin\left(\dfrac{1}{h}\right)$ oscillates between -1 and 1, the limit exists and equals 0 if and only if $m - 1 > 0 \implies m > 1$.

The smallest positive integer m is $m = 2$.

Now, for the cone:

Base radius $R = m = 2$

Height $H = 3m = 6$

$\dfrac{r}{R} = \dfrac{H - h}{H} \implies \dfrac{r}{2} = \dfrac{6 - h}{6} \implies h = 6 - 3r$

Volume of the cylinder $V(r) = \pi r^2 h = \pi r^2(6 - 3r) = 3\pi(2r^2 - r^3)$, where $0 < r < 2$.

$\dfrac{dV}{dr} = 3\pi(4r - 3r^2) = 0 \implies r = \dfrac{4}{3}$

Also, $\left.\dfrac{d^2V}{dr^2}\right|_{r=4/3} = 3\pi(4 - 6r)|_{r=4/3} = -12\pi < 0$, which confirms a local maximum.

$$V_{\max} = 3\pi \left[2\left(\frac{4}{3}\right)^2 - \left(\frac{4}{3}\right)^3 \right]$$

$$= 3\pi \left(\frac{32}{9} - \frac{64}{27} \right) = \frac{32\pi}{9}$$

Q13. If the function $f(x) = x^3 - 3x^2 - 9x + 5$ is strictly decreasing in the interval (a, b), then using differentials, the approximate value of $\sqrt{b - a + 0.24}$ is equal to:

(A) 2.060 (B) 2.063 (C) 2.057 (D) 2.054

Ans. A

Sol. $f'(x) = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$

For $f(x)$ to be strictly decreasing, $f'(x) < 0 \Rightarrow (x - 3)(x + 1) < 0$

$\Rightarrow x \in (-1, 3)$.

Thus, $a = -1$ and $b = 3$.

Hence, $b - a = 3 - (-1) = 4$.

We need to approximate $g(4.24)$ where $g(x) = \sqrt{x}$.

Let $x = 4$ and $\Delta x = 0.24$.

$g(x + \Delta x) \approx g(x) + g'(x)\Delta x$

$$\sqrt{4.24} \approx \sqrt{4} + \frac{1}{2\sqrt{4}}(0.24)$$

$$= 2 + \frac{0.24}{4} = 2 + 0.06 = 2.06$$

Q14. Let $f(x)$ and $g(x)$ be two real-valued functions defined as:

$$f(x) = \begin{cases} x^3 + 3x, & x \geq 1 \\ 3x^2 + 1, & x < 1 \end{cases} \text{ and } g(x) = \begin{cases} e^{x-1} + x, & x \geq 1 \\ x^2 + 1, & x < 1 \end{cases}$$

Then, the derivative of $f(x)$ with respect to $g(x)$ at $x = 1$ is:

(A) 6 (B) 2 (C) 3 (D) Does not exist

Ans. C

Sol. For $f(x)$:

$\lim\limits_{x \to 1^+} f(x) = f(1) = 1^3 + 3(1) = 4$

$\lim\limits_{x \to 1^-} f(x) = 3(1)^2 + 1 = 4$

Hence, $f(x)$ is continuous at $x = 1$.

Right hand derivative of $f(x)$ at $x = 1$: $Rf'(1) = \dfrac{d}{dx}(x^3 + 3x)\Big|_{x=1} = 3(1)^2 + 3 = 6$

Left hand derivative of $f(x)$ at $x = 1$: $Lf'(1) = \dfrac{d}{dx}(3x^2 + 1)\Big|_{x=1} = 6(1) = 6$

Since $Rf'(1) = Lf'(1) = 6$, $f(x)$ is differentiable at $x = 1$ and $f'(1) = 6$.

For $g(x)$:

$\lim\limits_{x \to 1^+} g(x) = g(1) = e^0 + 1 = 2$

$\lim_{x \to 1^-} g(x) = 1^2 + 1 = 2$

Hence, $g(x)$ is continuous at $x = 1$.

Right hand derivative of $g(x)$ at $x = 1$: $Rg'(1) = \dfrac{d}{dx}(e^{x-1} + x)\Big|_{x=1} = e^0 + 1 = 2$

Left hand derivative of $g(x)$ at $x = 1$: $Lg'(1) = \dfrac{d}{dx}(x^2 + 1)\Big|_{x=1} = 2(1) = 2$

Since $Rg'(1) = Lg'(1) = 2$, $g(x)$ is differentiable at $x = 1$ and $g'(1) = 2$.

Now, the derivative of $f(x)$ with respect to $g(x)$ at $x = 1$ is:
$\dfrac{df}{dg}\Big|_{x=1} = \dfrac{f'(1)}{g'(1)} = \dfrac{6}{2} = 3$

Q15. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \min\{2 - x^2, |x|\}$, and let $g(x) = \lfloor f(x) \rfloor$, where $\lfloor \cdot \rfloor$ denotes the greatest integer function. Then at $x = 1$, the function $g(x)$ has:

(A) a removable discontinuity (B) a jump (non-removable) discontinuity

(C) an essential (infinite) discontinuity (D) continuity

Ans. A

Sol. To analyze the behavior of $g(x) = \lfloor f(x) \rfloor$ at $x = 1$, we first evaluate $f(x) = \min\{2 - x^2, |x|\}$ in the neighborhood of $x = 1$:

1. For $x \in (0, 1)$, $|x| = x < 2 - x^2$, so $f(x) = x$.

Thus, for $x \in (0, 1)$, $0 < f(x) < 1 \implies g(x) = \lfloor f(x) \rfloor = 0$.

$\lim_{x \to 1^-} g(x) = 0$

2. For $x \in (1, \sqrt{2})$, $2 - x^2 < |x| = x$, so $f(x) = 2 - x^2$.

As $x \to 1^+$, $2 - x^2$ approaches 1 from values strictly less than 1.

Thus, for $x \in (1, \sqrt{2})$, $0 < f(x) < 1 \implies g(x) = \lfloor f(x) \rfloor = 0$.

$\lim_{x \to 1^+} g(x) = 0$

3. At $x = 1$: $f(1) = \min\{2 - 1^2, |1|\} = 1 \implies g(1) = \lfloor 1 \rfloor = 1$

Since $\lim_{x \to 1} g(x) = 0$ exists finitely, but $\lim_{x \to 1} g(x) \neq g(1)$, the function $g(x)$ has a **removable discontinuity** at $x = 1$.

Q16. Let C be the curve defined by $y = x \ln x$ for $x > 0$. For any point $P(x, y)$ on C with $x \neq \dfrac{1}{e}$, let $g(x)$ denote the y-intercept of the normal to the curve C at P. Define the function $f : (0, \infty) \to \mathbb{R}$ as:

$$f(x) = \begin{cases} g(x), & x \neq \frac{1}{e} \\ 0, & x = \frac{1}{e} \end{cases}$$

Then at $x = \dfrac{1}{e}$, the function $f(x)$ has:

(A) a removable discontinuity (B) a non-removable jump discontinuity

(C) an essential (infinite) discontinuity (D) no discontinuity (it is continuous)

Ans. C

Sol. For the curve $y = x \ln x$, we have:

$\frac{dy}{dx} = 1 + \ln x$

$m_N = -\frac{1}{1+\ln x}$

$Y - x\ln x = -\frac{1}{1+\ln x}(X - x)$

To find the y-intercept $g(x)$, set $X = 0$:

$g(x) = x\ln x + \frac{x}{1+\ln x}$

Now, consider the behavior of $f(x) = g(x)$ as $x \to \frac{1}{e}$:

As $x \to (\frac{1}{e})^+$, $\ln x > -1 \implies 1 + \ln x \to 0^+$, so $\frac{x}{1+\ln x} \to +\infty$.

Thus, $\lim\limits_{x\to(1/e)^+} f(x) = +\infty$.

Similarly, as $x \to (\frac{1}{e})^-$, $\ln x < -1 \implies 1 + \ln x \to 0^-$, so $\frac{x}{1+\ln x} \to -\infty$.

Thus, $\lim\limits_{x\to(1/e)^-} f(x) = -\infty$.

Since the one-sided limits do not exist finitely (they approach $\pm\infty$), the function $f(x)$ possesses an essential (infinite) discontinuity at $x = \frac{1}{e}$.

Q17. Let $f(x) = x^{\frac{1}{x}}$ for $x > 0$. If $L = \lim\limits_{x\to\infty} f(x)$ and $y(x) = \ln(f(x))$, then the value of $L + e^3 y''(e)$ is equal to:

(A) 0 (B) 1 (C) -1 (D) e

Ans. A

Sol. $\ln L = \lim\limits_{x\to\infty} \frac{\ln x}{x}$

$\ln L = \lim\limits_{x\to\infty} \frac{\frac{1}{x}}{1} = 0 \implies L = e^0 = 1$

Now, we have $y(x) = \ln(f(x)) = \frac{\ln x}{x}$.

$y'(x) = \frac{\frac{1}{x}\cdot x - \ln x \cdot 1}{x^2} = \frac{1-\ln x}{x^2}$

$y''(x) = \frac{\left(-\frac{1}{x}\right)x^2 - (1-\ln x)(2x)}{x^4}$

$= \frac{-x - 2x + 2x\ln x}{x^4} = \frac{2\ln x - 3}{x^3}$

$y''(e) = \frac{2\ln e - 3}{e^3} = \frac{2(1)-3}{e^3} = -\frac{1}{e^3}$

$L + e^3 y''(e) = 1 + e^3\left(-\frac{1}{e^3}\right) = 1 - 1 = 0$

Q18. Let $f(x)=\begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$, where p is a real constant. Let $g(x)=\tan^{-1}x$. If the derivative of $f(x)$ with respect to $g(x)$ at $x=0$ is equal to $f''(0)$, then the sum of all possible real values of p is:

(A) 0 (B) 1 (C) -1 (D) 2

Ans. B

Sol. $f'(x)=\begin{vmatrix} 3x^2 & \cos x & -\sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$

$f''(x)=\begin{vmatrix} 6x & -\sin x & -\cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$

At $x=0$:

$f'(0)=\begin{vmatrix} 0 & 1 & 0 \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}=-6p^3$

$f''(0)=\begin{vmatrix} 0 & 0 & -1 \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}=-1(6p^2+p)=-6p^2-p$

Now, $g(x)=\tan^{-1}x \implies g'(x)=\dfrac{1}{1+x^2} \implies g'(0)=1.$

The derivative of $f(x)$ with respect to $g(x)$ at $x=0$ is:

$\left.\dfrac{df}{dg}\right|_{x=0}=\dfrac{f'(0)}{g'(0)}=\dfrac{-6p^3}{1}=-6p^3$

Given that $\left.\dfrac{df}{dg}\right|_{x=0}=f''(0)$:

$-6p^3=-6p^2-p \implies 6p^3-6p^2-p=0$

$p(6p^2-6p-1)=0$

The real roots for p are $p=0$ and the roots of $6p^2-6p-1=0$, which are real since Discriminant $D=36-4(6)(-1)=60>0$.

The sum of all real values of p is $0+\left(-\dfrac{-6}{6}\right)=1.$

Q19. If a curve is given parametrically by the equations $x=2t^2+3$ and $y=4t^3-t$, where t is a real parameter, then the value of $\dfrac{dy}{dx}$ at $t=1$ is equal to :

(A) $\dfrac{11}{4}$ (B) $\dfrac{13}{4}$ (C) $\dfrac{9}{4}$ (D) $\dfrac{5}{2}$

Ans. A

Sol. $x=2t^2+3 \quad \text{and} \quad y=4t^3-t$

$\dfrac{dx}{dt}=4t$

$\dfrac{dy}{dt}=12t^2-1$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{12t^2 - 1}{4t}$$

Evaluating at $t = 1$:

$$\left.\frac{dy}{dx}\right|_{t=1} = \frac{12(1)^2 - 1}{4(1)} = \frac{11}{4}$$

Q20. If the function $f(x) = \frac{x^3}{3} - ax^2 + 4x + 1$ is strictly increasing on $\mathbb{R}$, and a_0 is the maximum non-negative integer value of a, then the length of the subnormal to the curve $y = f(x)$ at $x = 1$ for $a = a_0$ is:

(A) $\frac{5}{3}$ (B) $\frac{10}{3}$ (C) $\frac{20}{3}$ (D) 10

Ans. B

Sol. For $f(x)$ to be strictly increasing on $\mathbb{R}$, $f'(x) \geq 0$ for all $x \in \mathbb{R}$.

$f'(x) = x^2 - 2ax + 4 \geq 0$

Since the coefficient of x^2 is positive, we must have discriminant $D \leq 0$:

$(-2a)^2 - 4(1)(4) \leqslant 0 \Rightarrow 4a^2 - 16 \leqslant 0$

$\Rightarrow a^2 \leqslant 4 \Rightarrow -2 \leqslant a \leqslant 2$

The maximum non-negative integer value of a is $a_0 = 2$.

For $a = 2$, the function becomes:

$f(x) = \frac{x^3}{3} - 2x^2 + 4x + 1$

At $x = 1$:

$y = f(1) = \frac{1}{3} - 2 + 4 + 1 = \frac{10}{3}$

$f'(1) = 1^2 - 4(1) + 4 = 1$

The length of the subnormal is given by:

Length of subnormal $= |y \cdot f'(x)| = \left|\frac{10}{3} \cdot 1\right| = \frac{10}{3}$

Hence, the correct option is (B).

Q1. Let a curve be defined parametrically by $x = 2t$ and $y = t^2 + 1$, where $t \in \mathbb{R}$. If the shortest distance between the point $P(0, 5)$ and any point on the curve is d, then the value of d^2 is equal to _____.

Ans. 12

Sol. $\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = 2t \implies \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = t$

The slope of the tangent at Q is $m_T = t$, so the slope of the normal at Q is $m_N = -\frac{1}{t}$ (for $t \neq 0$).

The shortest distance from a point $P(0,5)$ to the curve occurs along the normal to the curve passing through P.

Equation of the normal at $Q(2t, t^2+1)$: $y-(t^2+1) = -\frac{1}{t}(x-2t)$

Since this normal passes through $P(0,5)$: $5-(t^2+1) = -\frac{1}{t}(0-2t)$ $\implies 4-t^2=2 \implies t^2=2$

Thus, the square of the shortest distance d^2 is: $d^2 = (2t-0)^2+(t^2+1-5)^2 = 4t^2+(t^2-4)^2$

Substituting $t^2=2$: $d^2 = 4(2)+(2-4)^2 = 8+4 = 12$

Q2. Let $f:(0,\infty)\to\mathbb{R}$ be defined by $f(x)=x^x$. If the value of the limit $\lim\limits_{x\to1}\dfrac{f(x)-x}{x-1-\ln x}$ is equal to L, then the value of L is _____.

Ans. 2

Sol. Given $f(x)=x^x$.

$\ln(f(x)) = x\ln x$

$\dfrac{f'(x)}{f(x)} = 1+\ln x \implies f'(x) = x^x(1+\ln x)$

Also, $f''(x) = x^x(1+\ln x)^2 + x^{x-1}$.

At $x=1$, we have $f(1)=1$, $f'(1)=1$, and $f''(1)=2$.

$L = \lim\limits_{x\to1}\dfrac{x^x-x}{x-1-\ln x}$

$L = \lim\limits_{x\to1}\dfrac{x^x(1+\ln x)-1}{1-\frac{1}{x}}$ As $x\to1$, this limit is again of the form $\dfrac{0}{0}$.

$L = \lim\limits_{x\to1}\dfrac{x^x(1+\ln x)^2+x^{x-1}}{\frac{1}{x^2}} = \dfrac{1(1+0)^2+1}{1} = 2$

Q3. Let $f:\mathbb{R}\to\mathbb{R}$ be a continuous function satisfying $f(x+y)=f(x)+f(y)+3xy(x+y)$ for all $x,y\in\mathbb{R}$, with $f'(0)=2$.

Consider the non-uniformly defined function $g:\mathbb{R}\to\mathbb{R}$ given by

$$g(x) = \begin{cases} f(x), & x\le 1 \\ \alpha x^2+\beta x+1, & x>1 \end{cases}$$

If $g(x)$ is differentiable at $x=1$, then the value of $\alpha-\beta$ is equal to ____.

Ans. 4

Sol. $f'(x) = \lim\limits_{h\to0}\dfrac{f(x+h)-f(x)}{h}$

$f(x+h) = f(x)+f(h)+3xh(x+h)$:

$f'(x) = \lim\limits_{h\to0}\dfrac{f(h)+3xh(x+h)}{h} = \lim\limits_{h\to0}\dfrac{f(h)}{h} + \lim\limits_{h\to0}3x(x+h) = f'(0)+3x^2$

$f'(x) = 3x^2 + 2$

$f(0) = 0$ gives: $f(x) = x^3 + 2x$

Now, for $g(x)$ to be differentiable at $x = 1$, it must first be continuous at $x = 1$:

$\lim\limits_{x \to 1^-} g(x) = \lim\limits_{x \to 1^+} g(x) = g(1)$

$f(1) = \alpha(1)^2 + \beta(1) + 1 \Rightarrow 1^3 + 2(1) = \alpha + \beta + 1 \Rightarrow \alpha + \beta = 2$ .....(i)

$g'(1^-) = f'(1) = 3(1)^2 + 2 = 5$

$g'(1^+) = \dfrac{d}{dx}(\alpha x^2 + \beta x + 1)\Big|_{x=1} = 2\alpha + \beta$

Since $g(x)$ is differentiable at $x = 1$: $2\alpha + \beta = 5$ --- (ii)

Solving equations (i) and (ii):$\alpha = 3, \quad \beta = -1$

Thus, $\alpha - \beta = 3 - (-1) = 4$.

Q4. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \begin{cases} x^m \cos\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ where $m \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Let A denote the number of values of m for which $f(x)$ is differentiable at $x = 0$, and let B denote the number of values of m for which $f'(x)$ is continuous at $x = 0$. Find the value of $A + B$ _____.

Ans. 17

Sol. $f'(0) = \lim\limits_{h \to 0} \dfrac{f(h) - f(0)}{h} = \lim\limits_{h \to 0} \dfrac{h^m \cos\left(\frac{1}{h}\right) - 0}{h} = \lim\limits_{h \to 0} h^{m-1} \cos\left(\dfrac{1}{h}\right)$

This limit exists and equals 0 if and only if $m - 1 > 0 \implies m > 1$.

Since $m \in \{1, 2, \ldots, 10\}$, $f(x)$ is differentiable at $x = 0$ for $m \in \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Thus, $A = 9$. For $x \neq 0$, the derivative is: $f'(x) = mx^{m-1} \cos\left(\dfrac{1}{x}\right) + x^{m-2} \sin\left(\dfrac{1}{x}\right)$

For $f'(x)$ to be continuous at $x = 0$, we must have $\lim\limits_{x \to 0} f'(x) = f'(0) = 0$.

This requires both $m - 1 > 0$ and $m - 2 > 0 \implies m > 2$.

Thus, $f'(x)$ is continuous at $x = 0$ for $m \in \{3, 4, 5, 6, 7, 8, 9, 10\}$, which gives $B = 8$.

Hence, $A + B = 9 + 8 = 17$.

Q5. Let $f(x) = x^4 - 4x^3 + 4x^2 + 10$. Let (a, b) be the open interval of maximum length contained in $(0, \infty)$ in which the function $f(x)$ is strictly decreasing. If $f''(x)$ denotes the second derivative of $f(x)$, then the value of $f''(a) + f''(b)$ is equal to _____.

Ans. 4

Sol. $f(x) = x^4 - 4x^3 + 4x^2 + 10$

$f'(x) = 4x^3 - 12x^2 + 8x = 4x(x^2 - 3x + 2) = 4x(x - 1)(x - 2)$

For $x > 0$, $f'(x) < 0$ when $(x - 1)(x - 2) < 0$, which gives $x \in (1, 2)$.
Thus, the required interval is $(a, b) = (1, 2)$, so $a = 1$ and $b = 2$.

$f''(x) = \frac{d}{dx}(4x^3 - 12x^2 + 8x) = 12x^2 - 24x + 8$

$a = 1$ and $b = 2$:

$f''(1) = 12(1)^2 - 24(1) + 8 = -4$

$f''(2) = 12(2)^2 - 24(2) + 8 = 8$

Therefore, $f''(a) + f''(b) = f''(1) + f''(2) = -4 + 8 = 4$